

SIMATS
ENGINEERING

SIMATS Online Programming Lab

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10/ 10 completed

Fib series-iterative & recur

FIB SERIES-ITERATIVE & RECUR

Write a C program to generate Fibonacci series up to N terms using both iterative and recursive methods. The program should display both outputs and compare execution time in milliseconds. Also display total recursive calls made.

Sample Input:

Enter N: 7

Sample Output:

Iterative: 0 1 1 2 3 5 8

Recursive: 0 1 1 2 3 5 8

Recursive Calls: 13

Iterative Time: 0.001 ms

Recursive Time: 0.005 ms

Test Case:

Enter N: 1

Output:

0

PROGRAM EDITOR

C

Run

Save

```
1 #include <stdio.h>
2 #include <time.h>
3
4 long recursiveCalls = 0;
5
6
7 void fibIterative(int n, int arr[]) {
8     int a = 0, b = 1, next;
9     for (int i = 0; i < n; i++) {
10         if (i == 0) {
11             arr[i] = a;
12         } else if (i == 1) {
13             arr[i] = b;
14         } else {
15             next = a + b;
16             a = b;
17             b = next;
18             arr[i] = next;
19         }
20     }
21 }
```

OUTPUT

```
Enter N: Iterative: 0 1 1 2 3 5 8
Recursive: 0 1 1 2 3 5 8
Recursive Calls: 59
Iterative Time: 0.000 ms
Recursive Time: 0.000 ms
```

INPUT

7

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